Finale

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Logistics

Assignment / Exam Grading

Likely to finish by end of week (Fri/Sat) so folks know scores going into the final Exam

Final Exam

- Thu Thu 14-May in ESJ 2208 (normal classroom)
- 10:30am - 12:30pm
- Comprehensive exam: all topics in the course eligible
- 6 sides of paper with problems broaching our areas of focus (see later slides for review topics)

Further Coursework

- **Application areas such as machine learning and data analysis** to see places you might properly apply your parallel skills
  - CMSC422 Introduction to Machine Learning
  - CMSC426 Computer Vision
  - CMSC454 Algorithms for Data Science
  - CMSC470 Introduction to Natural Language Processing
  - CMSC472 Introduction to Deep Learning
- **Science of Computation** to deepen your understanding of fundamental techniques in computation, often arising in HPC
  - CMSC460 Computational Methods
  - CMSC466 Introduction to Numerical Analysis I
  - CMSC660 Scientific Computing I
- **Hardware** to get further knowledge of the machine that we rely on, be it serial or parallel
  - CMSC411 Computer Systems Architecture
  - CMSC662 Computer Organization and Programming for Scientific Computing

Review Topic: Shared Memory Coordination

- Nearby is a pair of routines which sum random numbers in a sequence
- On running the code and timing with increasing numbers of threads, the following are observed

#threads	niters	wall time(s)	correct?
serial	10mil	8.05	Yes
omp 1	10mil	8.07	Yes
omp 2	10mil	12.32	No
omp 4	10mil	19.71	No
omp 8	10mil	30.68	No

Explain these and how to fix the parallel version

```
long randsum_serial(int niters){
  long sum = 0;
  for(int i=0; i<niters; i++){
    long num = rand();
    sum += num;
  }
  return sum;
}

long randsum_omp(int niters){
  long sum = 0;
  #omp parallel
  {
    for(int i=0; i<niters; i++){
      long num = rand();
      sum += num;
    }
  }
  return sum;
}
```

Review Topic: GPU Thread Sync

A

We have seen that CUDA provides the `__synchthreads()` function to synchronize threads within a thread block.
Why is this function necessary? Give an example where it would be useful.

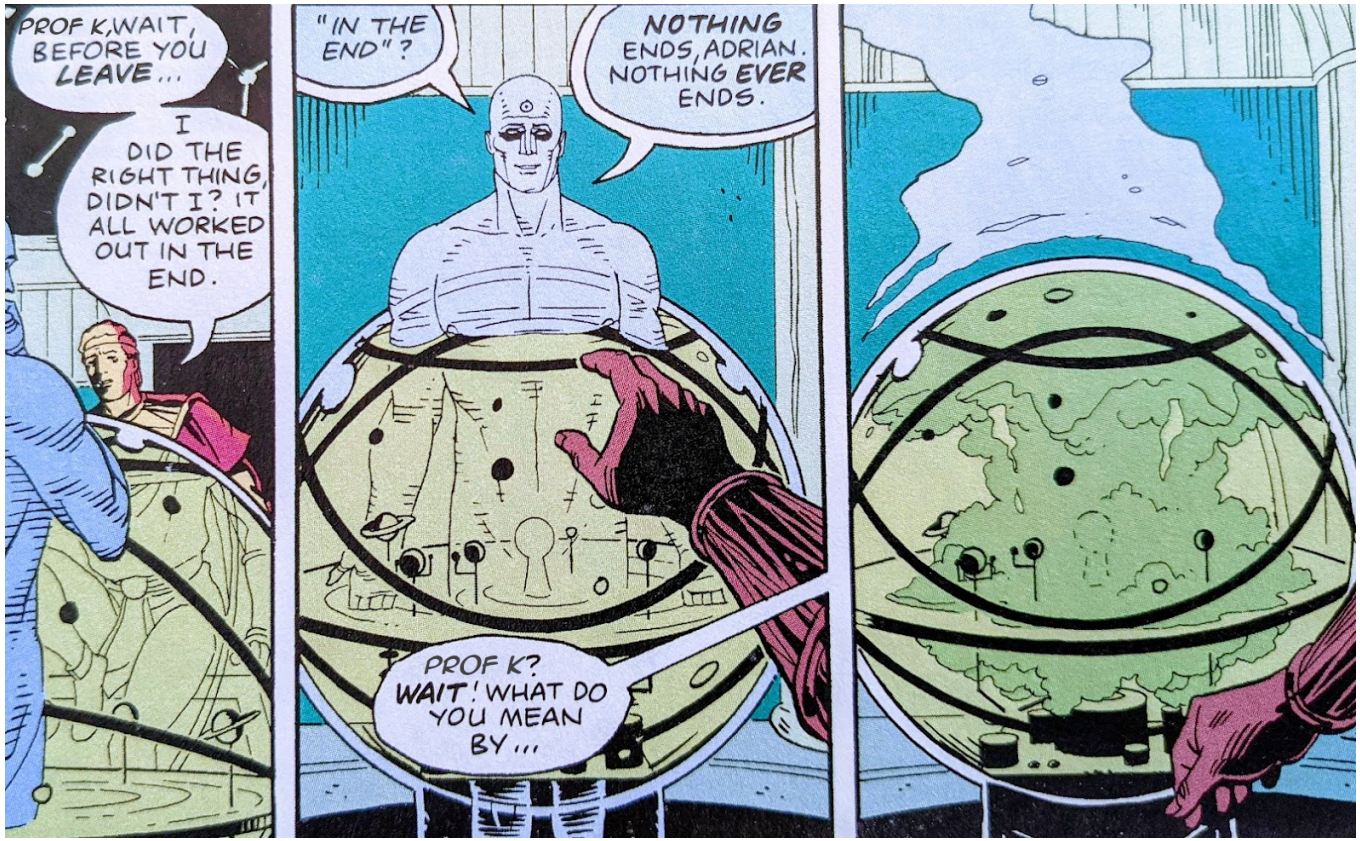
B

What are the limits to `__syncthreads()`?
Which threads does it synchronize and which threads does it not?
What alternatives are available for thread synchronization?

Review Topic Requests

- Distributed Memory Architecture
- Shared Memory Architecture
- MPI for Distributed Memory Programming
- PThreads + OpenMP for Shared Memory Programming
- CUDA for GPU Programming
- Communication Patterns
  - Broadcast, Scatter/Gather, Reductions
  - Synchronization between procs/threads
- Input/Output Problem Decomposition
- Theoretical and Practical Tools for Performance Analysis
- Parallel Problems/Applications
  - Sums, Min/Maxes
  - Matrix Multiplication
  - Solving Linear Systems
  - Sorting
  - Basics of N-Body simulation, Neural Networks, Crypto Mining, Heat Propagation, K-Means

Nothing Ever Ends



By now you should realize that what you learned

- Will come up again showing whether you learned it well the first time or need another pass.
- Will change in the future and make you feel old.

Expect this and stay stay patient.

Conclusion

It's been a hell of a semester.
I'm proud of all of you.
Keep up the good work.
Stay safe. Happy Hacking.